Troubleshooting Tools

Important: tasks 1 to 4 must be run on your own computer, so the answers depend on your network. Run each command, paste your own screenshot in the box, and fill in the blank. The 'sample output' images only show what the result looks like; their numbers are made up.

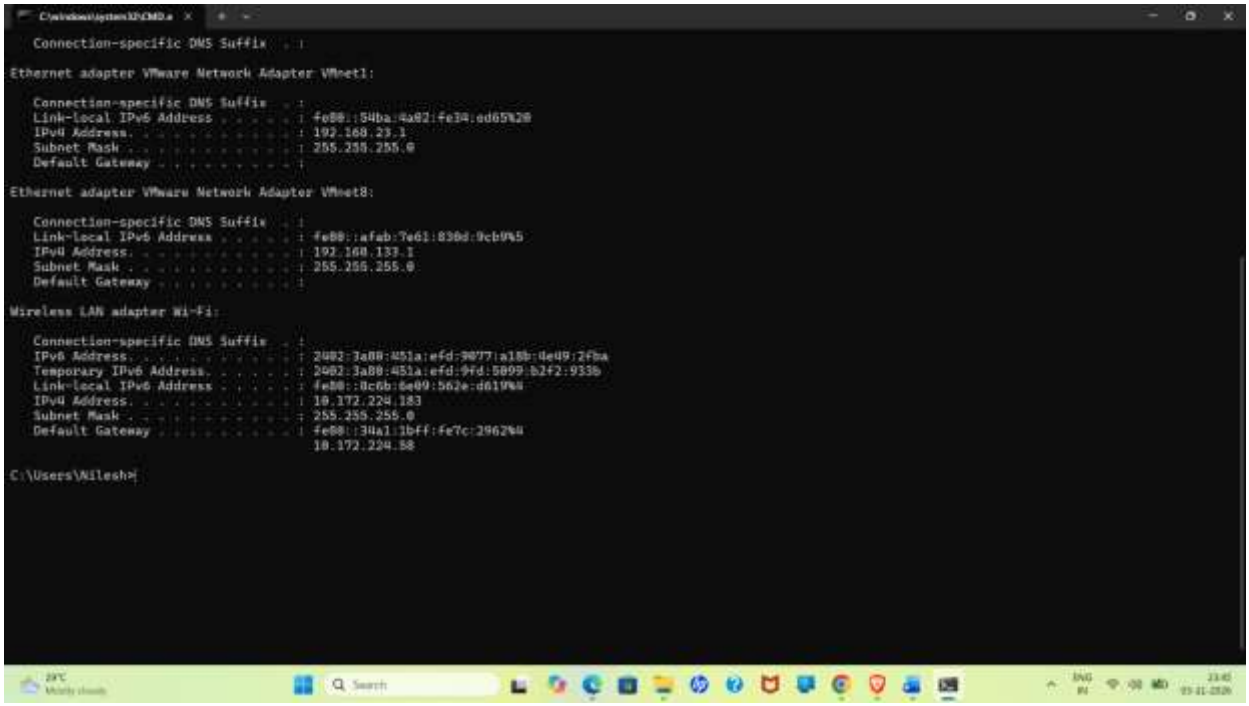
Task 1: ipconfig (IPv4 address and default gateway)

Open Command Prompt (press Windows + R, type cmd, press Enter) and run ipconfig. Look for 'IPv4 Address' and 'Default Gateway' under your active adapter.

Command to run:

```
ipconfig
```

Your screenshot (paste your own output here):



My IPv4 address: _____ My default gateway: _____

Sample output (example format only, with made-up values):

Command Prompt (SAMPLE OUTPUT)

```
C:\Users\Student> ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.1.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
```

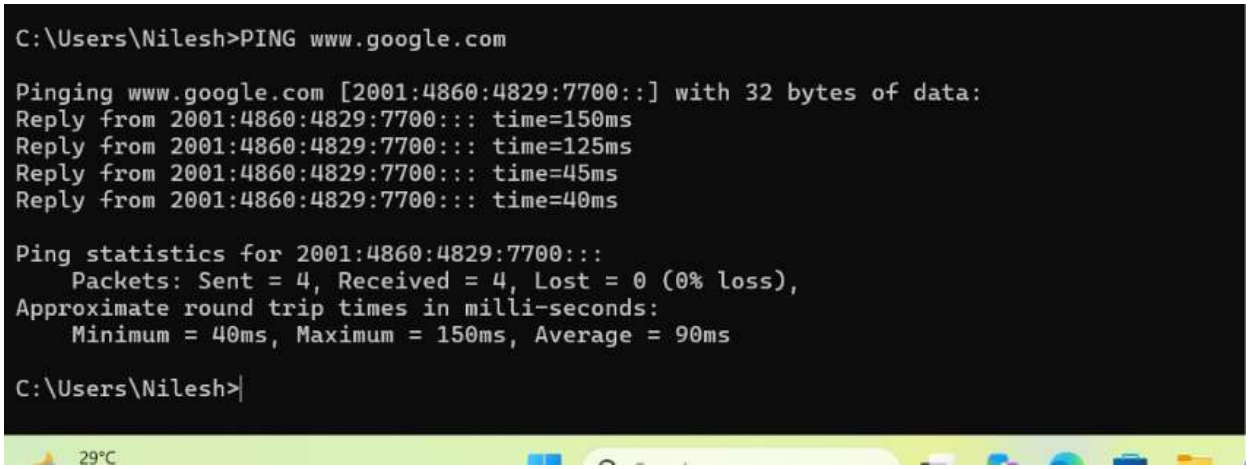
Task 2: ping www.google.com (average response time)

Run the ping command and read the 'Average' time at the bottom of the statistics.

Command to run:

```
ping www.google.com
```

Your screenshot (paste your own output here):



Average response time: _____ ms Packets lost: _____

Sample output (example format only, with made-up values):

Command Prompt (SAMPLE OUTPUT)

```
C:\Users\Student> ping www.google.com

Pinging www.google.com [203.0.113.10] with 32 bytes of data:
Reply from 203.0.113.10: bytes=32 time=22ms TTL=117
Reply from 203.0.113.10: bytes=32 time=25ms TTL=117
Reply from 203.0.113.10: bytes=32 time=24ms TTL=117
Reply from 203.0.113.10: bytes=32 time=25ms TTL=117

Ping statistics for 203.0.113.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 22ms, Maximum = 25ms, Average = 24ms
```

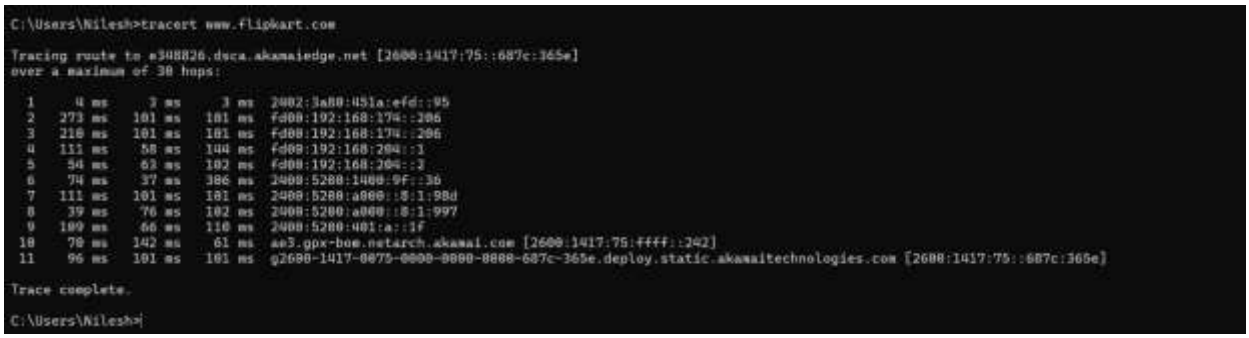
Task 3: tracert www.flipkart.com (number of hops)

Run tracert (Windows) or traceroute www.flipkart.com (Linux/Mac). Count the numbered lines before 'Trace complete'; the last number is the hop count. Rows showing '* * *' (Request timed out) are still hops, they are just routers that do not reply.

Command to run:

```
tracert www.flipkart.com
```

Your screenshot (paste your own output here):



Number of hops to reach the destination: _____

Sample output (example format only, with made-up values):

Command Prompt (SAMPLE OUTPUT)

```
C:\Users\Student> tracert www.flipkart.com

Tracing route to www.flipkart.com [198.51.100.20]
over a maximum of 30 hops:

  1     2 ms     1 ms     2 ms  192.168.1.1
  2     8 ms     7 ms     8 ms  100.64.0.1
  3    10 ms     9 ms    10 ms  203.0.113.1
  4    14 ms    13 ms    14 ms  203.0.113.45
  5    18 ms    17 ms    18 ms  198.51.100.5
  6    20 ms    21 ms    20 ms  198.51.100.9
  7    22 ms    22 ms    23 ms  198.51.100.20

Trace complete.
```

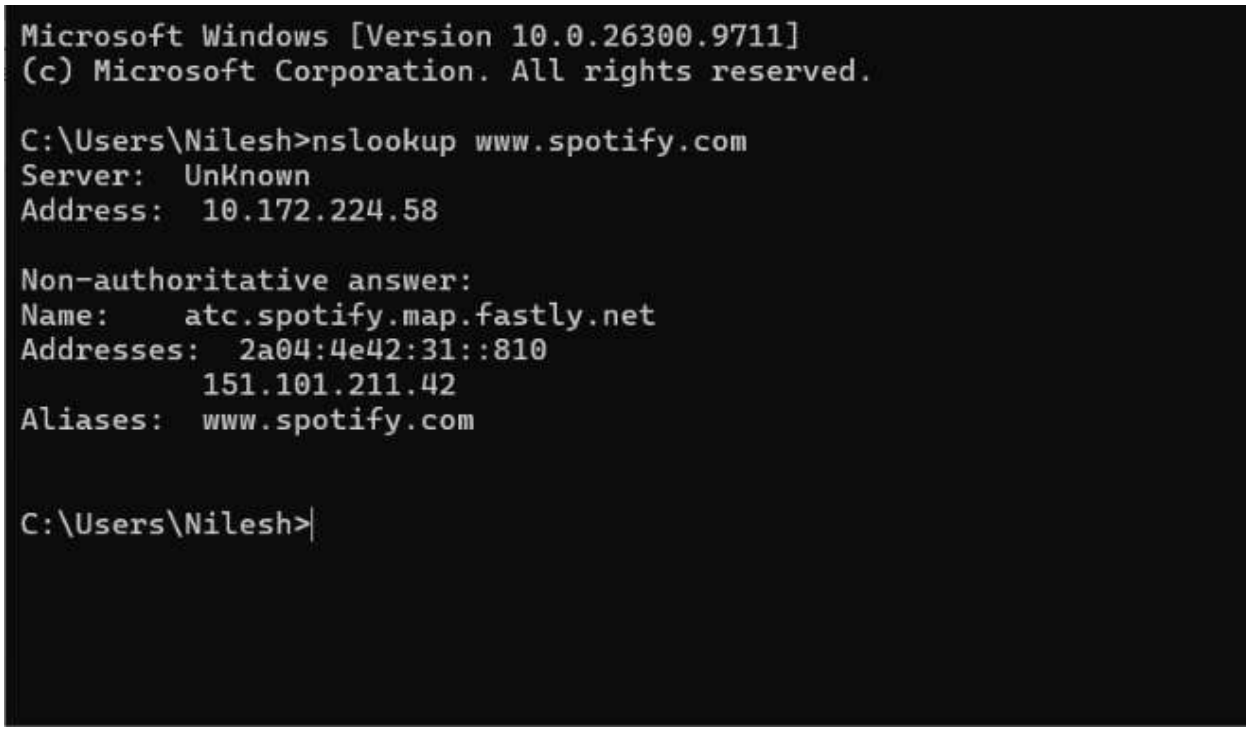
Task 4: nslookup www.spotify.com (IP address)

Run nslookup and note the address(es) listed under 'Name: www.spotify.com'. Ignore the 'Server' and 'Address' lines at the top; those belong to your DNS server.

Command to run:

```
nslookup www.spotify.com
```

Your screenshot (paste your own output here):



IP address of www.spotify.com: _____

Sample output (example format only, with made-up values):

Command Prompt (SAMPLE OUTPUT)

```
C:\Users\Student> nslookup www.spotify.com

Server:  dns.example.local
Address:  192.168.1.1

Non-authoritative answer:
Name:    www.spotify.com
Address:  198.51.100.77
```

One-line explanation:

nslookup helps you verify that DNS is working, that is, that a website name (www.spotify.com) is correctly converted into an IP address.

Task 5: Friend Cannot Open Zomato's Website

Run these three commands in this order, from the laptop's own network setup, to the internet, and then to the name lookup:

Command Prompt - three commands to run, in order

```
C:\> ipconfig
C:\> ping 8.8.8.8
C:\> nslookup www.zomato.com
```

Step	Command	What it checks	How to read the result
1	ipconfig	Whether the laptop has a valid network setup: an IPv4 address, subnet mask and default gateway.	If the IP starts with 169.254 or there is no gateway, the laptop is not connected properly to the WiFi/router. Fix this first.
2	ping 8.8.8.8	Whether the laptop can reach the internet by IP address, without using DNS.	Replies mean connectivity is fine. 'Request timed out' means the problem is the connection, the router or the ISP.
3	nslookup www.zomato.com	Whether DNS can turn the website name into an IP address.	An IP address in the answer means DNS works. 'Can't find' or 'timed out' means a DNS problem, so try changing the DNS server (for example to 8.8.8.8).

Conclusion:

If ipconfig is fine, ping 8.8.8.8 works, but nslookup fails, the problem is DNS. If ping 8.8.8.8 fails, the problem is the network connection. If all three work but the site still does not open, the problem is likely the browser, a firewall/proxy or the website itself. In that case, tracert www.zomato.com can show where the path stops.